

Machine Learning–based Differentiation of Benign and Premalignant Colorectal Polyps Detected with CT Colonography in an Asymptomatic Screening Population: A Proof-of-Concept Study

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Background: CT colonography does not enable definite differentiation between benign and premalignant colorectal polyps.

Purpose: To perform machine learning–based differentiation of benign and premalignant colorectal polyps detected with CT colonography in an average-risk asymptomatic colorectal cancer screening sample with external validation using radiomics.

Materials and Methods: In this secondary analysis of a prospective trial, colorectal polyps of all size categories and morphologies were manually segmented on CT colonographic images and were classified as benign (hyperplastic polyp or regular mucosa) or premalignant (adenoma) according to the histopathologic reference standard. Quantitative image features characterizing shape ($n = 14$), gray level histogram statistics ($n = 18$), and image texture ($n = 68$) were extracted from segmentations after applying 22 image filters, resulting in 1906 feature-filter combinations. Based on these features, a random forest classification algorithm was trained to predict the individual polyp character. Diagnostic performance was validated in an external test set.

Results: The random forest model was fitted using a training set consisting of 107 colorectal polyps in 63 patients (mean age, 63 years $\pm$ 8 [standard deviation]; 40 men) comprising 169 segmentations on CT colonographic images. The external test set included 77 polyps in 59 patients comprising 118 segmentations. Random forest analysis yielded an area under the receiver operating characteristic curve of 0.91 (95% CI: 0.85, 0.96), a sensitivity of 82% (65 of 79) (95% CI: 74%, 91%), and a specificity of 85% (33 of 39) (95% CI: 72%, 95%) in the external test set. In two subgroup analyses of the external test set, the area under the receiver operating characteristic curve was 0.87 in the size category of 6–9 mm and 0.90 in the size category of 10 mm or larger. The most important image feature for decision making (relative importance of 3.7%) was quantifying first-order gray level histogram statistics.

Conclusion: In this proof-of-concept study, machine learning–based image analysis enabled noninvasive differentiation of benign and premalignant colorectal polyps with CT colonography.

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In industrialized countries, colorectal cancer is among the three most common causes of cancer-related death (1,2). It is assumed that most types of colorectal cancer originate from adenomatous polyps developing over several years (3). Thus, the incidence and mortality of colorectal cancer can be reduced by early detection of precancerous polyps with consecutive resection (4–6). As clinical symptoms are unspecific and often absent, particularly in the early stages, screening procedures such as immunochemical fecal occult blood test and optical colonoscopy (OC) play a major role in colorectal cancer prevention (7,8).

During the past 2 decades, CT colonography emerged as a noninvasive screening method for colorectal cancer. The sensitivity of CT colonography and OC for the detection of colon polyps that are 6 mm or larger (and hence advanced adenoma detection rate)

are comparable in asymptomatic screening populations and in patients with symptoms suggestive of colorectal cancer (9–11). Also, CT colonography is effective in visualizing portions of the colon not evaluated by OC in cases of complex anatomic conditions causing failed or incomplete OC and therefore permits robust polyp detection also in the right colon (12). But CT colonography does not enable a definite differentiation between benign and premalignant polyps, crucial for individual risk stratification and therapy guidance. Therefore, polyp size measured in CT colonography data sets is currently used as surrogate indicator of the likelihood of malignancy. Current guidelines recommend OC-guided resection for colorectal polyps that are 6 mm or larger (United States Multi-Society Task Force on Colorectal Cancer, European Society of Gastrointestinal

Abbreviations

AUC = area under the receiver operating characteristic curve, OC = optical colonoscopy

Summary

In this proof-of-concept study, machine learning–assisted CT colonography analysis allowed for differentiation of benign and premalignant colorectal polyps at high diagnostic performance associated with the histopathologic reference standard.

Key Results

- Radiomics-based image analysis enables noninvasive differentiation of benign and premalignant CT colonography–detected colorectal polyps, with an area under the receiver operating characteristic curve of 0.91, a sensitivity of 82%, and a specificity of 85%.
- The area under the receiver operating characteristic curve for the machine learning–based differentiation of benign versus premalignant polyps was 0.87 in the 6–9-mm size category and 0.90 in the 10-mm or larger size category.

Endoscopy, and European Society of Gastrointestinal and Abdominal Radiology) (13,14).

Radiomics describes the process of converting medical image data sets into mineable high-dimensional data by extracting quantitative image features based on intensity, shape, size or volume, and texture. Using machine learning approaches, such as random forests, support vector machines, or neural networks, tumor-specific radiomics features enable comprehensive tumor characterization beyond the visible morphology in radiologic images (15–17). For instance, CT texture features predicted KRAS (Kirsten rat sarcoma viral oncogene homolog) mutation status of manifest colorectal cancer with an area under the receiver operating characteristic curve (AUC) of 0.83 (18).

The aim of this proof-of-concept study was to establish a noninvasive, radiomics-based machine-learning differentiation of benign (ie, hyperplastic polyp or regular mucosa) and premalignant (ie, adenoma) polyps in CT colonography data sets from an asymptomatic, average-risk colorectal cancer screening cohort.

Materials and Methods

Training Set

The present study was approved by the institutional review board, and the requirement for written informed consent was waived. The random forest machine learning model in this retrospective analysis was trained using CT colonography scans from a previously published prospective colorectal cancer screening cohort of an average-risk asymptomatic population older than 50 years undergoing fecal occult blood test, fecal immunochemical stool test, and same-day OC and CT colonography to compare the performance of these screening tests in the detection of advanced colonic neoplasia (9). Participants were enrolled in the previously published colorectal cancer screening cohort only if they had no symptoms of colonic diseases, such as melanic stools, hematochezia, abdominal pain, relevant changes in stool frequency, diarrhea, inflammatory bowel disease, hereditary colorectal can-

cer syndromes, positive family history for colorectal cancer (ie, one first-degree relative with colorectal cancer diagnosis before 60 years of age or two first-degree relatives with colorectal cancer diagnosis at any age), prior OC within the past 5 years, body weight greater than 150 kg, and severe cardiovascular or pulmonary disease (9). In the present study, a polyp-enriched data set was created, including only participants with histopathologically characterized polyps (Fig 1).

CT Colonography in the Training Set

Bowel preparation included 4 L of polyethylene glycol solution (KleanPrep; Norgine Pharmaceuticals) and a combination of 20 mg bisacodyl with 30 mL of sodium phosphate (Prepacol; Guerbet Pharma). A volume of 50 mL of the iodinated contrast agent iopamidol (Solutrast 300; BraccoAltana Pharma) was added to the last liter of polyethylene glycol electrolyte solution to tag residual fluid in the colon. Image data sets were acquired with a 64-channel multidetector row scanner (Somatom Sensation 64; Siemens Healthineers) at 0.6-mm collimation. Images were reconstructed at a section thickness of 0.75 mm and a 0.5-mm reconstruction increment using a standard soft-tissue kernel. CT scans in the supine position were acquired at a tube voltage of 120 kVp and tube current–time product reference values of 70 mAs. CT scans in the prone position were acquired at a tube voltage of 120 kVp and tube current–time product reference values of 30 mAs using an online dose modulation technique for automatic tube current adaption (Care Dose 4D; Siemens Healthineers). Dose-length products were recorded as estimates of radiation exposure. Mean radiation dose for CT colonography was 4.5 mSv (0.6). Bowel distention was achieved via manual air or automated carbon dioxide insufflation via a rectal tube and evaluated by a radiologist on the CT scout film of the abdomen. The first set of images was obtained in a 7–9-second breath hold in the supine position, and the second set was obtained after repositioning of the study participant in the prone position. Bowel preparation and the CT colonography protocol were previously described in detail (9). All bowels were adequately distended, cleansed, and tagged to ensure high diagnostic performance of CT colonography.

External Test Set

To estimate generalization performance of our machine learning model, we used external CT colonography data sets from a large North American multicenter CT colonography screening trial made publicly accessible via The Cancer Imaging Archive (19–21). The Cancer Imaging Archive is a large multicenter open-source open-access collection of anonymized medical images of cancer, including radiologic data sets. The CT colonography data sets of the aforementioned multicenter screening trial were acquired with multiple CT scanners from several vendors (Siemens Healthineers, GE Healthcare Systems, Philips Healthcare, Canon Medical Systems) with varying scanning protocols. Only polyps with available histopathologic reports were included. Polyps that could not be unequivocally identified because of poor bowel distension or insufficient

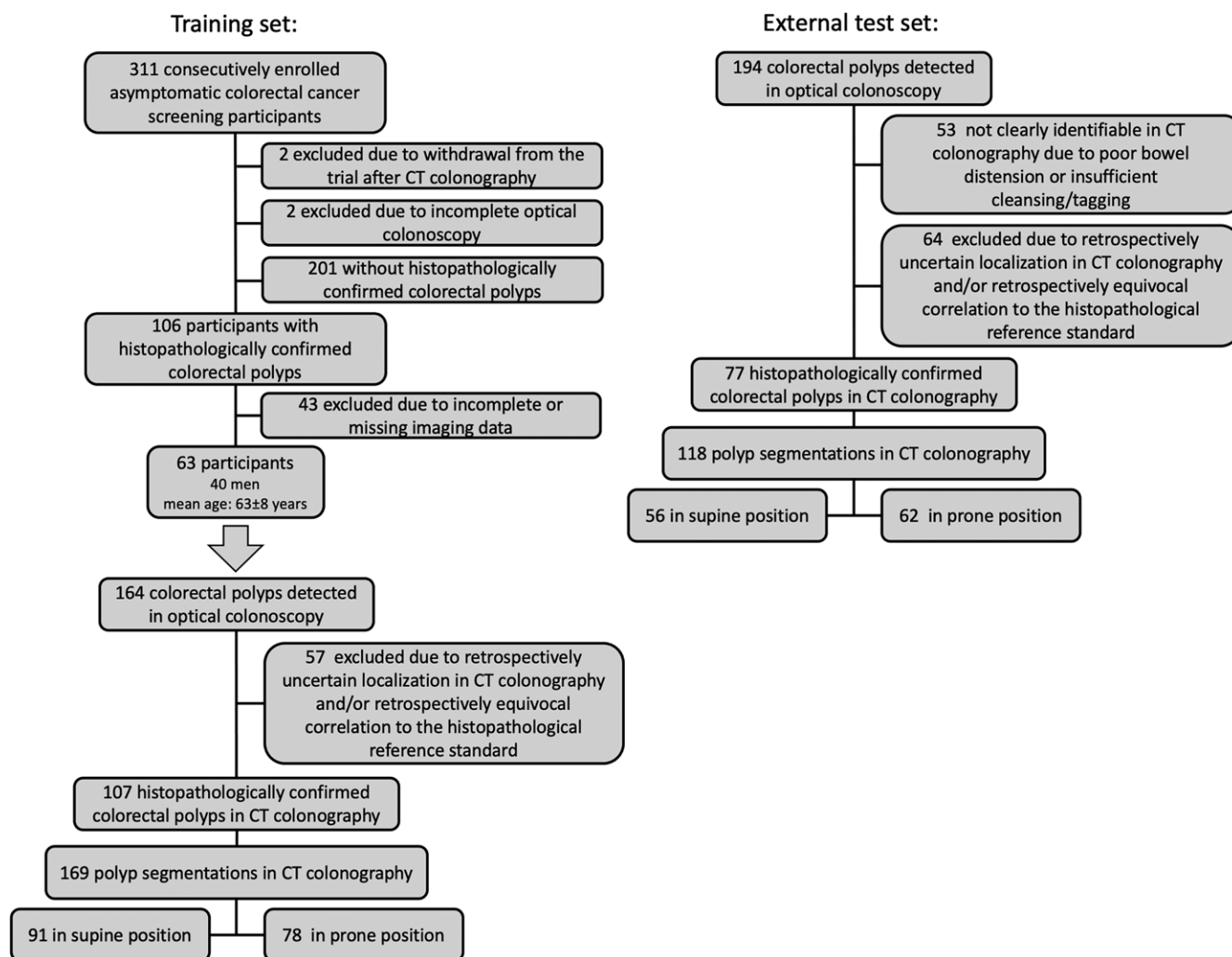


Figure 1: Flow diagram of the training set and external test set.

cleansing or tagging were excluded from the analysis. Polyps with retrospectively equivocal or uncertain correlation to the histopathologic reference standard were also excluded.

Image Segmentation

Colorectal polyps in both data sets were manually segmented for radiomics feature extraction by a board-certified radiologist (P.M.K., 8 years of experience in oncologic whole-body imaging) and two radiology residents (S.G., S.M.; 3 years of experience in oncologic whole-body imaging) in multiplanar two-dimensional CT colonography images (Fig 2). Information on polyp size and colon segment in which polypectomy was performed was provided to all readers. All readers were blinded to the histopathologic polyp class. Polyp segmentations were performed by each reader in equal amounts. Each polyp segmentation was confirmed by the other two readers who did not perform the segmentation. In case of divergent reading results, a consensus reading was performed. To assess intrareader, interreader, and radiomics feature variability, 25% of polyp segmentations from the test set were randomly selected with stratification by class label (benign or premalignant) and were again segmented independently by the same

reader (intrareader variability) and a second reader (interreader variability). These second polyp segmentations were performed 5 months after the first segmentations. Combined reading of two-dimensional CT colonography images and virtual fly-through three-dimensional reconstructions was used for exact polyp detection. Manual polyp segmentation was performed with multiplanar two-dimensional CT colonography images. Each colorectal polyp was segmented on supine and prone position scans if clearly identifiable in both positions. If a polyp could be located in only the supine or prone position, it was segmented in that position only. Polyps that could not be securely identified or unequivocally assigned to the corresponding histopathologic report were excluded from analysis. Polyp size categories were 5 mm or smaller, 6–9 mm, and 10 mm or larger. In addition, polyps were morphologically classified as pedunculated, sessile, or flat. The CT colonography workflow of the commercially available dedicated postprocessing software syngo.via version VA30B (Siemens Healthineers) was used for exact polyp detection and localization. The free open-source software Medical Imaging Interaction Toolkit, version 2018.04 (German Cancer Research Center) was used for manual segmentation (22).

Histopathologic Reference Standard

Colorectal polyps were included only if they were unequivocally assignable to the corresponding histopathologic report (Fig 3). A polyp was considered benign if it was diagnosed as a hyperplastic polyp or regular mucosa in the corresponding histopathologic report. A polyp was considered premalignant if it was diagnosed as tubular adenoma, tubulovillous adenoma, or villous adenoma in the corresponding histopathologic report. For study purposes only, a small number of lesions with the histopathologic classification serrated adenoma (four polyp segmentations) or adenocarcinoma (five segmentations) were included in the premalignant group. Correspondingly, two polyp segmentations with the histopathologic classification lipomatous polyp were categorized in the benign group.

Feature Extraction

Quantitative image features were extracted from segmented voxels in CT colonography scans using the open-source Python package Pyradiomics (version 2.2.0; Harvard Medical School) (23). Quantitative image features, including gray level histogram statistics ($n = 18$) and image texture ($n = 68$), were extracted after applying 22 image filters for each, and 14 characterizing shapes (including size) also were extracted, resulting in 1906 (ie, $[18 + 68] \cdot 22 + 14$) feature-filter combinations, which will be referred to simply as *features*. Feature extraction is illustrated in Figure 4 as part of the entire radiomics workflow. Lists of all image filters and extracted features are provided in Tables E3 and E4 (online). To facilitate feature reproducibility and comparability of polyps scanned at different centers, all CT colonography scans were preprocessed prior to feature extraction, including a resampling of CT pixel spacing to 0.72 mm along the x-axis and y-axis and 0.5 mm along the z-axis.

Feature Selection

Among the 1906 features previously extracted, features highly correlated with another feature were identified and excluded from the analysis. This feature selection based on pairwise feature correlation was applied to improve the machine learning training process and to enable optimized feature interpretability.

For this purpose, we calculated a Pearson correlation matrix for all features extracted from the training set and excluded one feature of any feature pair ($1906 \times 1905/2 = 1.8$ million pairs) with a Pearson correlation coefficient greater than 0.8. The exclusion of features that were too highly correlated reduced the number of features used for the final analysis to 10% (198 of 1906).

Random Forest Training

All 169 polyp segmentations in the training set were class-divided according to the histopathologic reference standard into benign or premalignant. The random forest classifier `sklearn.ensemble.RandomForestClassifier` (Python Scikit-learn machine learning library, version 0.22 [24]) with 1000 trees ($n_estimators = 1000$) and otherwise default parameters was trained on the 169 polyp segmentations of the training set using the previously identified set of 198 features to predict polyp class (ie, to differentiate between benign and premalignant colorectal polyps). Each decision tree in the random forest was trained on bootstrap resamples of the entire training data. Inside the random forest tree, binary decisions

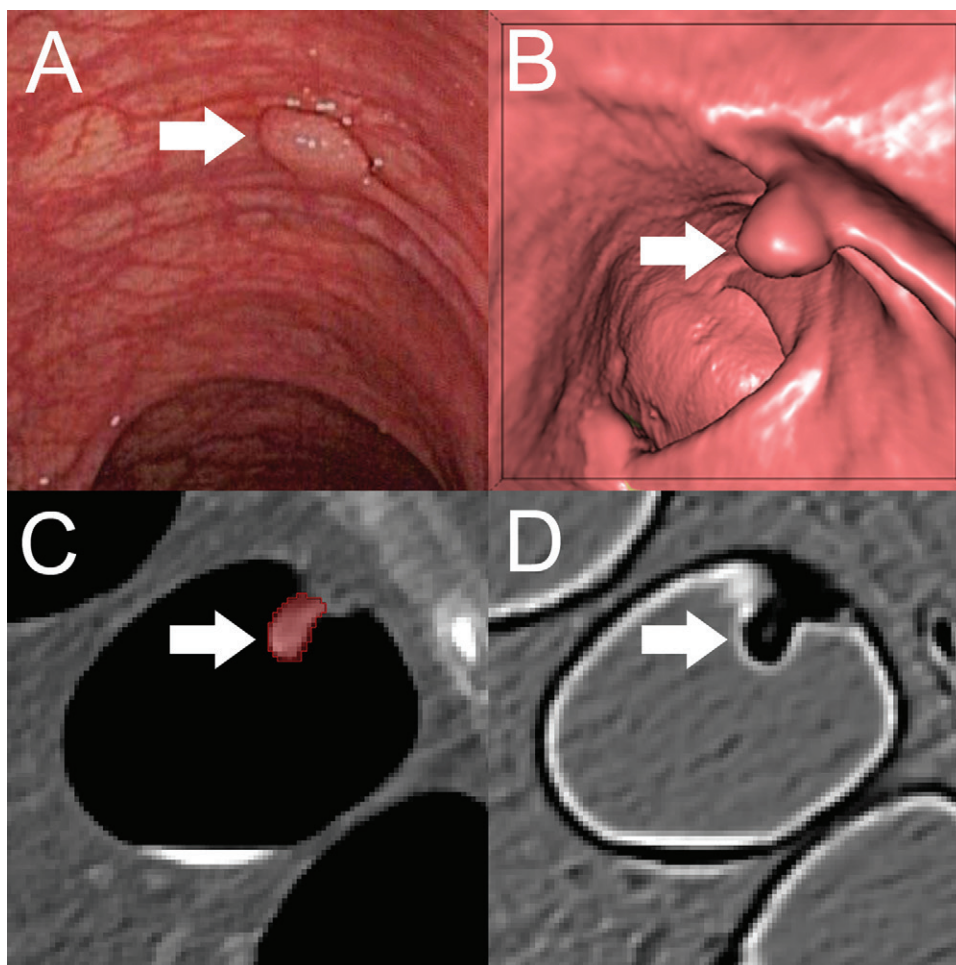


Figure 2: A, Optical colonoscopy and, B–D, CT colonography of a 9-mm polyp (arrow) in the descending colon of a 78-year-old woman. B, Virtual fly-through three-dimensional reconstructions were used for exact polyp localization. C, Manual polyp segmentation was performed in multiplanar two-dimensional CT colonography images. D, CT colonography images were preprocessed for image feature extraction by application of a dedicated filter.

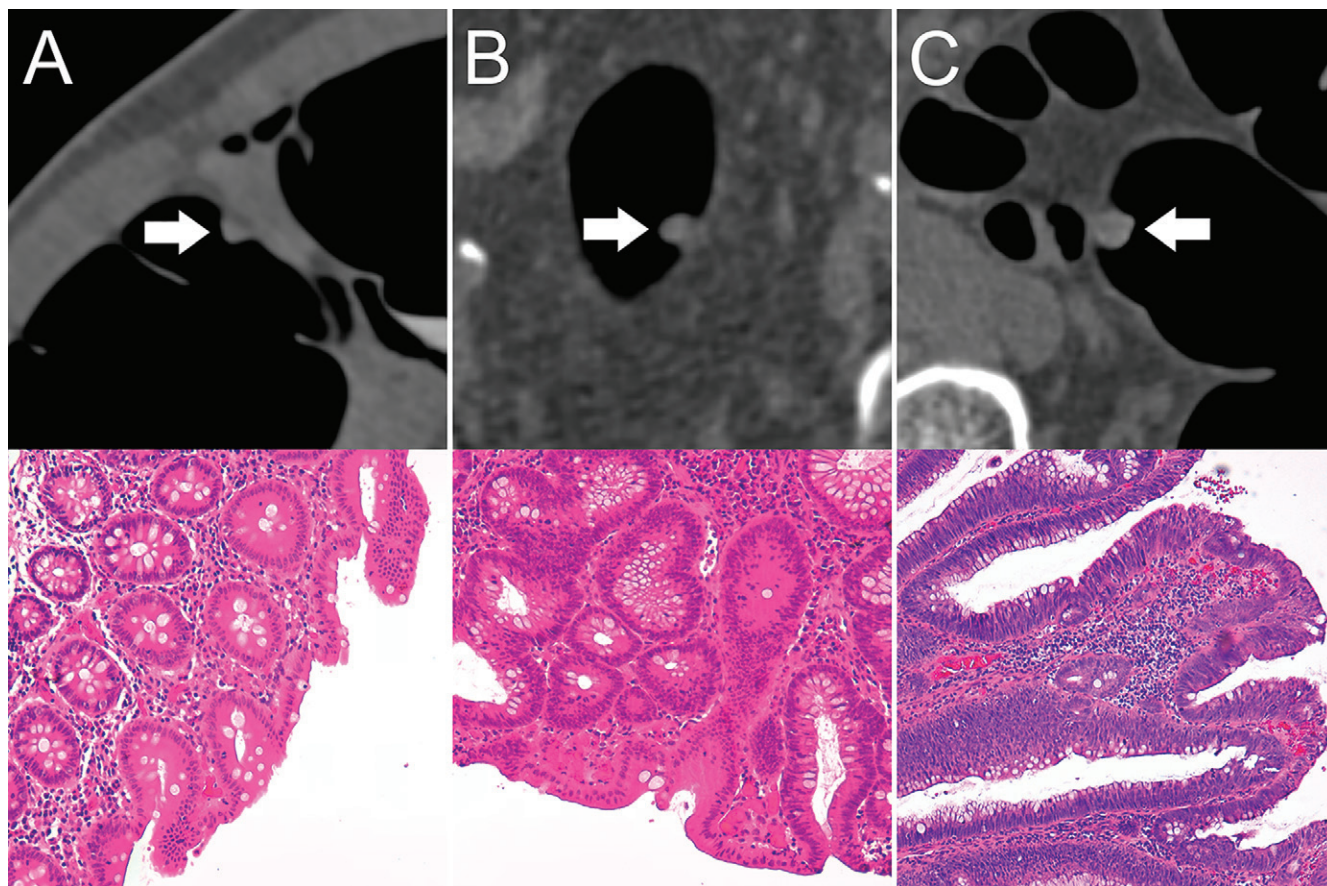


Figure 3: Top: Axial CT colonography images show representative colorectal polyps (arrow) in the training set. Bottom: Corresponding histopathologic work-up. (Hematoxylin-eosin staining; original magnification $\times 20$.) A, An 8-mm hyperplastic polyp in the ascending colon of a 54-year-old woman with hyperplastic epithelia. B, An 8-mm tubular adenoma in the sigmoid colon of a 68-year-old man with tubular growth pattern and elongated nuclei. C, An 11-mm tubulovillous adenoma in the rectum of a 73-year-old man with tubulovillous growth pattern and elongated nuclei.

at individual nodes were learned on randomly chosen feature subsets. This twofold randomness served to grow decision trees that are independent to the largest extent possible, such that “the generalization error converges almost surely to a limit as the number of trees becomes large” (25). The Scikit-learn random forest implementation followed Breiman et al (25) with one minor exception: It combined classifiers by averaging their probabilistic prediction instead of letting each classifier vote for a single class. Compared with other popular machine learning algorithms (eg, AdaBoost), random forests are robust to outliers and noise (25,26). Additionally, random forests provide internal error estimates that can also be used for estimating feature importance.

Statistical Analysis of the Test Set

Diagnostic performance of our machine learning–based polyp classification approach was assessed by evaluating the previously trained random forest on the external test set. Therefore, the 118 polyp segmentations in the external test set were class-divided into benign and premalignant in the same manner as the training set, and the same set of 198 features used for training was extracted per image. Performance was quantified with three metrics: AUC, sensitivity, and specificity. Sensitivity and specificity depended on the classifica-

tion threshold, a parameter that was used to turn predicted class probabilities, the output of the random forest model for a given input sample, into class predictions (benign vs premalignant). Sensitivity and specificity were evaluated for a default threshold value of 0.5 (model A), a threshold value that maximized the Youden index ($J = \text{sensitivity} + \text{specificity} - 1$) (27) (model B), and a threshold that resulted in the highest possible specificity while achieving a sensitivity of at least 85% (model C). For the AUC, sensitivity, and specificity, 95% CIs were calculated from 2000 bootstrap samples of the external test set (28). The random forest analysis is shown in Figure 5.

Statistical Analysis of the Training Set and Model Introspection

Training set samples left out for training a tree, so-called out-of-bag samples, were used to self-evaluate the respective tree and ultimately to form an internal prediction error estimate for the random forest, which is called the out-of-bag error (29). In addition, the Scikit-learn random forest implementation provided an internal estimate of feature importance, namely how much the class prediction (benign vs premalignant) of a trained model depended on a specific feature relative to all other features. We exploited this by evaluating the relative importance

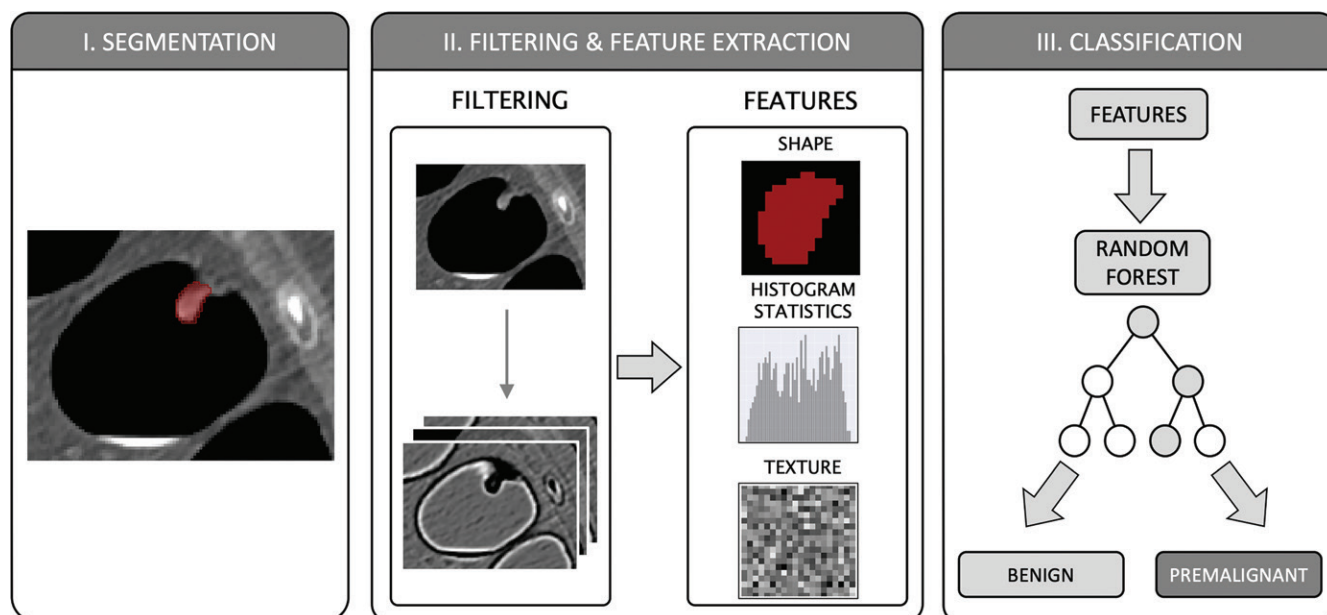


Figure 4: The radiomics workflow comprised three steps: manual segmentation of colorectal polyps in multiplanar two-dimensional CT colonography images; image filtering and feature extraction characterizing shape, histogram statistics, or texture; and feature-based training of a random forest classification algorithm to differentiate between benign and premalignant colorectal polyps according to the histopathologic reference standard.

of the 198 features used for training and testing the random forest. A detailed explanation of feature importance assessment is described in Appendix E1 (online).

During all stages of the analysis, the training and external test data sets were kept separate. The machine learning models were trained on the training set and evaluated on the test set. The statistical analysis was implemented in Python, and the entire code was made publicly available on the development platform Github (<https://github.com/pwespl/random-forest-polyp-classification>).

Results

Training Set

Of 311 consecutively enrolled asymptomatic adults who underwent same-day CT colonography and OC, two (1%) had to be excluded because of withdrawal from the trial after CT colonography and two (1%) had to be excluded because of incomplete OC, as reported previously (9). In this retrospective analysis, 201 of 307 (65%) screening participants without histopathologically confirmed colorectal polyps were excluded. Of the resulting 106 colorectal cancer screening participants with histopathologically confirmed polyps, 43 (41%) were excluded because of incomplete or missing CT colonography imaging data. Of 164 colorectal polyps detected in OC, 57 (35%) were excluded because of retrospectively uncertain localization in CT colonography, retrospectively equivocal correlation to the histopathologic reference standard, or both. Thirty-five of the 57 (61%) excluded polyps were benign, 22 of 57 (39%) were premalignant, and further details are presented in Table E1 (online). Consensus reading due to divergent reading results was performed in five of 107 (5%) polyps. A total of 107 colorectal polyps were analyzed in 63 patients (mean age, 63 years $\pm$ 8; 40 men) comprising 169 segmentations of polyps

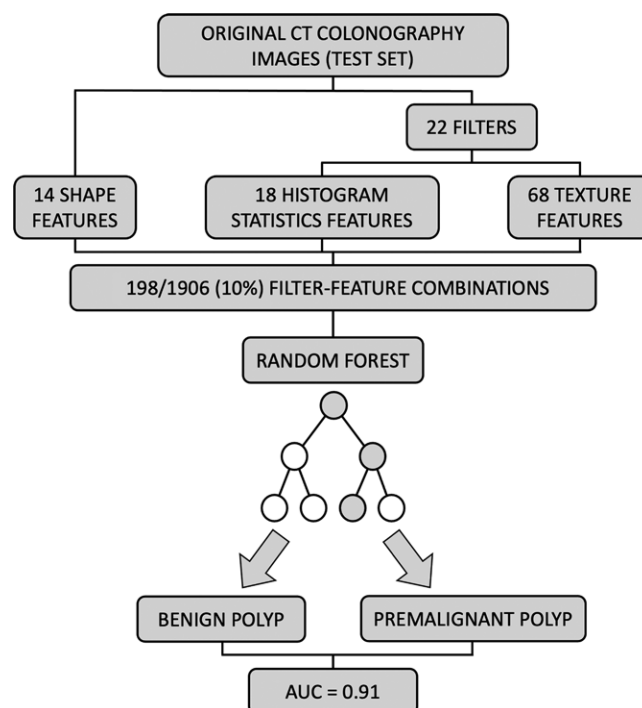


Figure 5: External validation of the trained random forest model. A total 198 of 1906 (10%) feature filter combinations were extracted from original CT colonography images of the test set using different image filters ($n = 22$) and image features characterizing shape ($n = 14$), histogram statistics ($n = 18$), or texture ($n = 68$). On the basis of these filter feature combinations, the trained random forest classifier was used to predict the colorectal polyp class label (benign vs premalignant). Prediction performance was quantified using area under the receiver operating characteristic curve (AUC).

(91 of 169 [54%] in the supine position, 78 of 169 [46%] in the prone position), as presented in Figure 1 and Table E2 (online). Of 169 polyp segmentations, 24 (14%) were 5 mm or smaller,

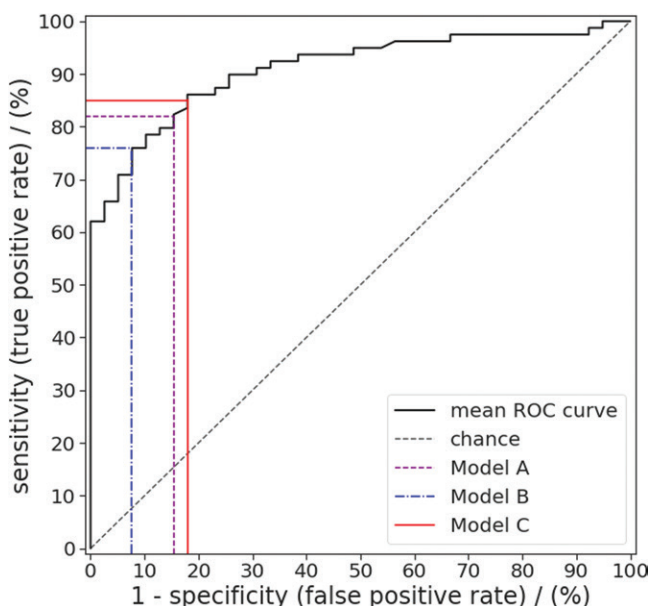


Figure 6: Receiver operating characteristic (ROC) curve for random forest predictions of colorectal polyp class (benign vs premalignant) for the polyps in the external test set. Sensitivity and specificity were evaluated for a default threshold value of 0.5 (model A), a threshold value that maximized the Youden index (model B), and a threshold resulting in the highest possible specificity while achieving a sensitivity of at least 85% (model C).

67 (40%) measured between 6 and 9 mm, and 78 of 169 (46%) were 10 mm or larger (maximum three-dimensional diameter measured through polyp segmentation). Eighty-three of 169 (49%) polyp segmentations were classified as benign (ie, hyperplastic polyp or regular mucosa), of which 16 of 83 (19%) were 5 mm or smaller, 49 of 83 (59%) were between 6 and 9 mm, and 18 of 83 (22%) were 10 mm or larger. Eighty-six of 169 (51%) polyp segmentations were classified as premalignant (ie, adenoma), of which eight of 86 (9%) were 5 mm or smaller, 18 of 86 (21%) were between 6 and 9 mm, and 60 of 86 (70%) were 10 mm or larger, as shown in Table 1. Polyp segmentation morphologies according to the size categories 5 mm or smaller, 6–9 mm, and 10 mm or larger are presented in Table 2.

External Test Set

A total of 53 of 194 (27%) colorectal polyps detected in OC were not clearly identifiable at CT colonography because of poor bowel distention or insufficient cleansing or tagging and were consequently excluded. Sixty-four of 194 (33%) polyps detected in OC were excluded because of retrospectively uncertain identification at CT colonography, equivocal correlation to the histopathologic reference standard, or both. A total of 58 of 117 (50%) excluded polyps were benign, whereas 59 of 117 (50%) were premalignant; further details are presented in Table E1 (online). Consensus reading due to divergent reading results was performed in five of 77 (7%) polyps. A total of 77 colorectal polyps was analyzed in 59 patients comprising 118 segmentations of polyps (56 of 118 [47%] in the supine position, 62 of 118 [53%] in the prone position). Nine of 118 (8%) polyp segmentations were 5 mm or smaller, 56 of 118

(47%) were between 6 and 9 mm, and 53 of 118 (45%) were 10 mm or larger, measuring the maximum three-dimensional diameter through polyp segmentation. Of 118, 39 (33%) polyp segmentations were classified as benign (ie, hyperplastic polyp or regular mucosa), of which eight (21%) were 5 mm or smaller, 26 (67%) were between 6 and 9 mm, and five (13%) were 10 mm or larger. Of the 118 polyp segmentations, 79 (67%) were classified as premalignant (ie, adenoma), of which one (1%) was 5 mm or smaller, 30 (38%) were between 6 and 9 mm, and 48 (61%) were 10 mm or larger, as shown in Table 1. Polyp segmentation morphologies according to the size categories 5 mm or smaller, 6–9 mm, and 10 mm or larger are presented in Table 2.

Statistical Analysis of the Test Set

Machine learning predictions of polyp class (benign vs premalignant) for all polyps in the external test set were derived from a random forest model fitted to the training set, yielding an overall AUC of 0.91 (95% CI: 0.85, 0.96). Sensitivity and specificity for a default threshold value of 0.5 (model A) were 82% (95% CI: 74, 91) (65 of 79) and 85% (95% CI: 72, 95) (33 of 39), respectively. Sensitivity and specificity for a threshold value of 0.53, which maximized the Youden index (model B), were 76% (95% CI: 64, 94) (60 of 79) and 92% (95% CI: 76, 100) (36 of 39), respectively. Sensitivity and specificity for a threshold value of 0.48, which resulted in the highest possible specificity while achieving a sensitivity of at least 85% (model C), were 85% (95% CI: 85, 89) (67 of 79) and 82% (95% CI: 61, 95) (32 of 39), respectively (Fig 6).

In subgroup analyses of the external test, which only contained polyps of a certain size category, machine learning predictions of polyp class (benign vs premalignant) yielded an AUC of 0.87 in the size category 6–9 mm and 0.90 in the size category 10 mm or larger. Because current guidelines recommend OC-guided resection for colorectal polyps 6 mm or larger (United States Multi-Society Task Force on Colorectal Cancer, European Society of Gastrointestinal Endoscopy, and European Society of Gastrointestinal and Abdominal Radiology) the number of polyps 5 mm or smaller with available histopathologic reports in the external test set (nine polyp segmentations) was not sufficient to provide a reliable AUC for this size category (13,14).

In a subgroup analysis of the external test set, in which only a single segmentation of each colorectal polyp was included, machine learning predictions of polyp class (benign vs premalignant) yielded an AUC of 0.90 for supine segmentations only and of 0.90 for prone segmentations only, respectively.

In an intrareader variability analysis, for which 29 of 118 (25%) polyp segmentations from the test set were segmented again by the same reader (Dice score, 0.80), random forest model predictions (benign vs premalignant) yielded an AUC of 0.87. In an interreader variability analysis, for which 29 of 118 (25%) polyp segmentations from the test set were segmented again by a second reader (Dice score, 0.77), random forest model predictions yielded an AUC of 0.91.

Table 1: Colorectal Polyp Segmentation Characteristics according to the Histopathologic Report of the Training Set and External Test Set

Histopathologic Category	No. of Polyp Segmentations		Classification
	Training Set (<i>n</i> = 169)	External Test Set (<i>n</i> = 118)	
Regular mucosa	3 (2)	9 (8)	Benign
Hyperplastic polyp	78 (46)	30 (25)	Benign
Lipomatous polyp	2 (1)	0 (0)	Benign
Tubular adenoma	57 (34)	49 (42)	Premalignant
Tubulovillous adenoma	16 (9)	26 (22)	Premalignant
Villous adenoma	8 (5)	0 (0)	Premalignant
Serrated adenoma	4 (2)	0 (0)	Premalignant
Adenocarcinoma	1 (1)	4 (3)	Premalignant

Note.—Data in parentheses are percentages. For study purposes only, a small number of adenocarcinoma segmentations were included in the premalignant group.

Table 2: Colorectal Polyp Segmentation Morphology in Three Size Categories of Both the Training Set and External Test Set

Morphologic Category	No. of Polyp Segmentations					
	Training Set			External Test Set		
	≤5 mm	6–9 mm	≥10 mm	≤5 mm	6–9 mm	≥10 mm
Pedunculated	0/169 (0)	5/169 (3)	32/169 (19)	0/118 (0)	1/118 (1)	17/118 (14)
Sessile	23/169 (14)	57/169 (34)	43/169 (25)	9/118 (8)	52/118 (44)	31/118 (26)
Flat	1/169 (1)	5/169 (3)	2/169 (1)	0/118 (0)	3/118 (3)	1/118 (1)
Carcinomatous	0/169 (0)	0/169 (0)	1/169 (1)	0/118 (0)	0/118 (0)	4/118 (3)

Note.—Data are proportions, and data in parentheses are percentages. The three size categories include the maximum three-dimensional diameter measured through polyp segmentation.

Statistical Analysis of the Training Set and Model Introspection

Polyp class predictions (benign vs premalignant) for out-of-bag training set samples yielded an AUC of 0.88. The 10 (of 198 [5%]) most important image features responsible for random forest classifications in this study were deduced from the trained random forest model and are shown in Table 3. Their relative feature importance added up to 23.8%. Seven features characterized texture (ranks 2, 3, 6–10), two features quantified first-order gray level histogram statistics (ranks 1 and 5), and one feature assessed size (rank 4). The remaining 188 (95%) features had a combined relative feature importance of 76.2%.

In detail, among the image features used for training the random forest model and making predictions, the size-measuring feature “original_shape_LeastAxisLength” was ranked as the fourth most important feature, with a relative feature importance of 2.6%, as shown in Table 3. It measured the smallest axis lengths of polyp segmentation-enclosing ellipsoids based on a principal component analysis.

Discussion

CT colonography does not enable a definite differentiation between benign and premalignant colorectal polyps, which would be crucial for individual risk stratification and therapy guidance. Consequently, in this proof-of-concept study, we investigated the noninvasive machine learning-based differentiation

of benign and premalignant polyps in CT colonography data sets of an asymptomatic average-risk colorectal cancer screening cohort over 50 years of age. In correlation to the histopathologic reference standard, machine learning-based image analysis enabled robust differentiation of benign and premalignant CT colonography-detected colorectal polyps with an area under the receiver operating characteristic curve (AUC) of 0.91, even for small polyps in the size category of 6–9 mm with an AUC of 0.87. External validation of the technique in a large North American multicenter trial patient cohort demonstrated robustness and reproducibility of our method, despite data sets acquired by vari-

ous scanner types and heterogeneous CT colonography imaging protocols (19–21). The feature importance analysis showed that a size-measuring image feature was in fourth place, but interestingly it was the only feature assessing size among the 10 most important image features for decision making in correlation to the histopathologic reference standard. The other nine image features characterized the distribution and texture of gray levels.

Our results are in line with pioneering studies by Aman et al and Song et al (30,31). Aman et al showed, in a data set of 97 polyps, that the differentiation of benign from premalignant polyps in machine learning-assisted CT colonography analysis using content-based image retrieval achieved a significantly higher ($P = .048$) AUC of 0.76 as opposed to the polyp size-only approach, with an AUC of 0.66 (30). Song et al reached an AUC of 0.85 for the machine learning-assisted differentiation of benign from premalignant colorectal polyps in a data set of 148 colorectal polyps, of which 35 were benign and 113 were premalignant (31). However, no external validation was performed to ensure the reliability in different scanner and protocol settings in these studies. In addition, Aman et al did not provide details on polyp size, and Song et al only included polyps 8 mm or larger (30,31). Our study adds to the field, as polyps smaller than 8 mm were analyzed, the machine learning algorithm was trained on a balanced training set, a standardized set of Pyradiomics image features was used, and validation on an external multicenter test set was performed.

Table 3: Ten Most Important Pyradiomics Image Features Responsible for Random Forest Model Classifications

Rank	Feature	Relative Importance (%)
1	log-sigma-2-0-mm-3D_firstorder_10Percentile	3.7
2	wavelet-LLH_glcmm_Idmn	3.3
3	square_gldm_SmallDependenceLowGrayLevelEmphasis	3.2
4	original_shape_LeastAxisLength	2.6
5	original_firstorder_90Percentile	2.4
6	original_glcmm_Imc2	2.2
7	wavelet-HLH_gldm_LargeDependenceHighGrayLevelEmphasis	1.9
8	wavelet-LLH_gldm_LongRunLowGrayLevelEmphasis	1.5
9	exponential_glszm_SmallAreaLowGrayLevelEmphasis	1.5
10	gradient_glcmm_Contrast	1.5

Note.—The relative feature importance added up to 23.8%. Seven features characterized texture (ranks 2, 3, 6–10), two features quantified first-order gray level histogram statistics (ranks 1 and 5), and one feature assessed size (rank 4). The remaining 188 (of 198 total) (95%) features had a combined relative feature importance of 76.2%. Filter abbreviations, feature category abbreviations, and feature abbreviations are presented in Tables E5–E7 (online).

Reproducibility between research groups and between image data sets acquired with different scanners using varying acquisition protocols pose a great challenge for machine learning–assisted image analysis (32,33). To address this issue and to ensure reproducibility of our machine learning model, only the standardized radiomics features available in the open-source Python package Pyradiomics were used (23), which is compliant with the Image Biomarker Standardization Initiative (34). The effect of intra- and interreader variability on our machine learning algorithm was investigated. To reduce the effect of varying CT colonography protocols, all CT colonography scans were resampled prior to feature extraction. Importantly, an external multicenter test set comprising CT colonography images acquired with multiple CT scanners from several vendors using varying CT colonography protocols was used to assess the performance of our radiomics-based machine learning differentiation of benign and premalignant colorectal polyps (19–21).

Brenner et al showed that OC cancer screening reduces the risk of colorectal cancer (7,35). However, these very successful OC screening programs had a low participation rate, at approximately 15%–20% (36–38). Colorectal cancer screening programs using noncathartic CT colonography and full cathartic preparation CT colonography show a higher participation rate compared with OC: 982 of 2920 (34%) for noncathartic CT colonography versus 1276 of 5924 (22%) for OC and 612 of 2430 (25%) for full cathartic preparation CT colonography versus 153 of 1036 (15%) for OC, respectively (37,38). Adding machine learning–assisted image analysis to conventional radiologic image reading could further improve the clinical importance of CT colonography–based colorectal cancer screening by allowing for a more precise selection of patients eligible for subsequent OC-guided polypectomy. Furthermore, machine learning–assisted CT colonography image analysis potentially provides the gastroenterologist with a road map of premalignant polyps eligible for OC-guided resection, particularly in patients

with a high burden of colorectal polyps, such as in familial adenomatous polyposis.

Our study had limitations. The sample size was small, with 107 histopathologically confirmed colorectal polyps in the training set and 77 histopathologically confirmed colorectal polyps in the external test set. Every polyp securely identifiable in CT colonography data sets and unequivocally assignable to the corresponding histopathologic report

was segmented. In the retrospective setting, however, a substantial number of polyps detected with OC did not meet these criteria and had to be excluded. Consequently, a selection bias could not be completely ruled out, and our results are only applicable to polyps clearly detectable with CT colonography. To ensure reproducibility of our machine learning model, only the standardized radiomics features available in the open-source Python package Pyradiomics were used (23). A dedicated feature measuring polyp height was not included, which might have been important for classification decisions, as hyperplastic polyps are known to be flatter than adenomatous polyps in CT colonography (39). Each patient, including all corresponding polyps and segmentations, was part of either the training set or the external test set. No patient who was presented to the random forest model during training was presented to the model again during testing. In a subgroup analysis of the external test set, in which only a single segmentation of each colorectal polyp was included, machine learning predictions of polyp class (benign vs premalignant) yielded an AUC of 0.90 for supine segmentations only and 0.90 for prone segmentations only. However, correlations within multiple polyps of one patient or within multiple segmentations of one polyp cannot be ruled out.

In this proof-of-concept study with validation in an external multicenter test set, machine learning–assisted CT colonography analysis enabled the differentiation of benign and premalignant colorectal polyps. The present study provides a potential basis for future prospective studies with a larger number of patients to further examine the diagnostic performance of machine learning algorithms for the noninvasive analysis of CT colonography–detected polyps.

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